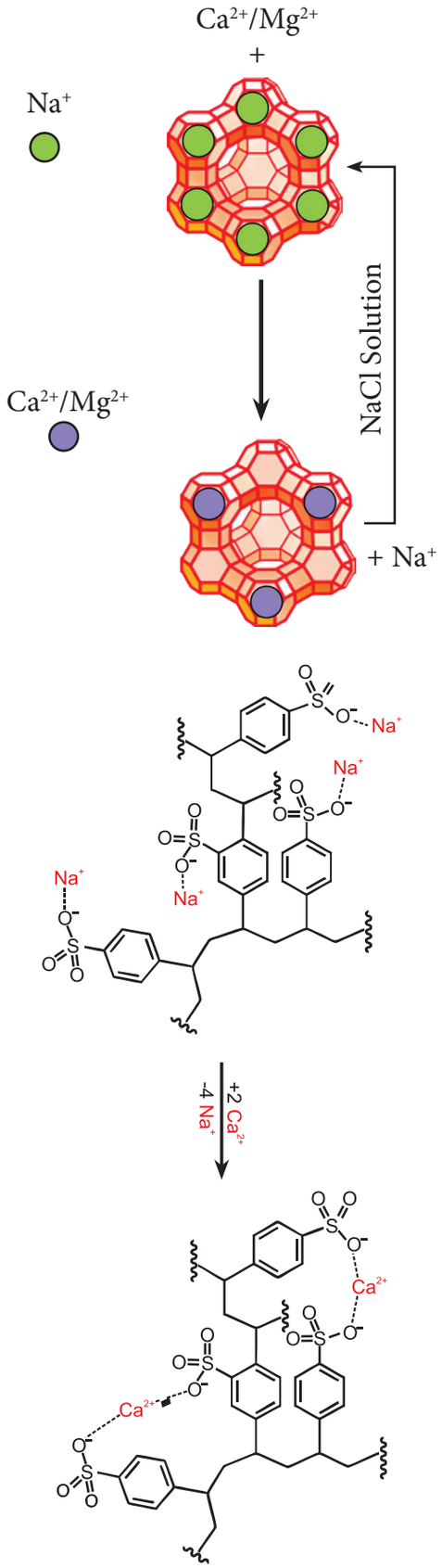


4.6 Heavy Water:

Heavy water (D_2O) is the oxide of heavy hydrogen. One part of heavy water is present in 5000 parts of ordinary water. It is mainly obtained as the product of electrolysis of water, as D_2O does not undergo electrolysis as easily as H_2O.

D_2O is a colorless, odorless and tasteless liquid. However, there is a marked difference between physical properties of water and heavy water as shown in Table 4.2.



Hard water produces less foam with detergents. Do you know why?

The cleaning capacity of soap is reduced when used in hard water. Soaps are sodium or potassium salts of long chain fatty acids (e.g., coconut oil). When soap is added to hard water, the divalent magnesium and calcium ions present in hard water react with soap. The sodium salts present in soaps are converted to their corresponding magnesium and calcium salts which are precipitated as scum/precipitate.

$$M^{2+} + 2RCOONa \rightarrow (RCOO)_2M + 2Na^+$$

M = Ca or Mg

$R = C_{17}H_{35}$

4.6.1 Chemical properties of heavy water:

When compounds containing hydrogen are treated with D_2O, hydrogen undergoes an exchange for deuterium

11th Std Chemistry 100-124.indd 110 06-01-2021 15:24:57